

VoIP SYSTEM INSTALLATION GUIDE

A Comprehensive Guide for ICT & Networking Students

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TVET CERTIFICATE III IN NETWORKING AND INTERNET TECHNOLOGIES

RWANDA TVET BOARD

At the end of the module the learner will be able to:

1. Plan VoIP system installation
2. Deploy VoIP system equipment
3. Configure VoIP system
4. Operate VoIP system
5. Test VoIP system
6. Maintain VoIP system

❖ IDENTIFICATION OF VOIP SYSTEM INSTALLATION REQUIREMENTS

A **VoIP system (Voice over Internet Protocol)** is a technology that allows people to **make voice calls using the Internet** instead of traditional telephone lines (PSTN).

VoIP (Voice over Internet Protocol) is a communication technology that allows people to **make voice calls using Internet Protocol (IP) networks**, such as the Internet or a local area network (LAN), instead of traditional telephone lines.

Traditional phone lines are the old, classic telephone systems that allow people to make voice calls **without using the Internet**. They use **physical copper wires** and circuit-switched technology.

These systems are also called **PSTN**.

PSTN = Public Switched Telephone Network

It is the **global system of traditional telephone networks** that connects landline phones worldwide.

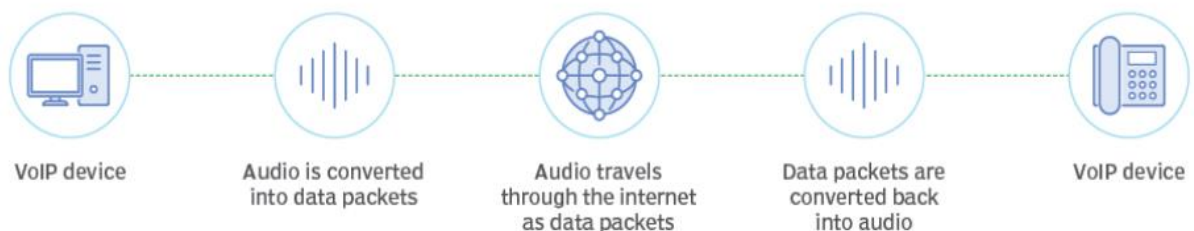
VoIP converts your **voice into digital data packets** and sends them over an **IP network** (such as the Internet or a local network). The data is then converted back into voice at the receiver's end.

Simply, **VoIP = Calling through the Internet**.

How a VoIP System Works (Simple Steps)

1. You speak into a VoIP phone or app
2. Your voice is **digitized and compressed**
3. Data packets are sent over the Internet
4. The receiver's device converts data back to sound

How VoIP works



❖ Types of VoIP Systems

VoIP (Voice over Internet Protocol) System Types

What is VoIP? (Quick Reminder)

VoIP is a technology that allows people to make **voice calls using the Internet** instead of traditional telephone lines.

📌 Example: Calling someone on WhatsApp using Wi-Fi instead of airtime.

1. Residential VoIP Services

Definition

Residential VoIP services are VoIP systems designed for **home and personal use**.

Examples

- Skype
- Google Voice
- Vonage
- MagicJack

Real-Life Analogy

Just like using **home Wi-Fi to watch YouTube**, residential VoIP lets you **call friends and family using the internet**.

Applications (Uses)

- Calling relatives abroad
- Online interviews
- Video and voice chats from home
- Saving money on international calls

2. Business VoIP Services

Definition

Business VoIP services are VoIP solutions designed for **small, medium, and large businesses** to manage professional communication.

Examples

- RingCentral
- 8x8

- Nextiva
- Jive
- Cisco Webex Calling

Real-Life Analogy

Like a **school office telephone system** where many staff members share lines and extensions.

Applications

- Customer support call centers
- Office communication
- Remote work and meetings
- Professional phone numbers for businesses

3. On-Premises VoIP Systems

Definition

On-premises VoIP systems are **installed and managed inside the organization** using their own servers and equipment.

Examples

- Asterisk
- FreePBX
- Cisco Unified Communications Manager

Real-Life Analogy

Like keeping your **own generator at school** instead of relying on public electricity.

Applications

- Large companies with IT teams
- Schools and universities
- Organizations needing full control of their system

4. Hosted (Cloud-Based) VoIP Systems

Definition

Hosted VoIP systems are **managed by a service provider on the internet**. The company does not maintain the servers themselves.

Examples

- Grasshopper
- Dialpad
- Ooma Office
- VirtualPBX

Real-Life Analogy

Like using **Google Drive** instead of storing files on your own computer.

Applications

- Small businesses
- Startups
- Companies without IT staff
- Easy and quick setup

5. Mobile VoIP Applications

Definition

Mobile VoIP apps allow users to **make calls on smartphones using mobile data or Wi-Fi**.

Examples

- WhatsApp
- Viber
- FaceTime
- Facebook Messenger

Real-Life Analogy

Just like sending a **WhatsApp message**, but instead you're making a voice or video call.

Applications

- Daily communication
- Group calls
- Free international calling
- Online classes and meetings

6. Softphones

Definition

Softphones are **software applications** that turn a computer or mobile device into a VoIP phone.

Examples

- Zoiper
- X-Lite
- Bria
- Linphone

Real-Life Analogy

Like using a **calculator app instead of a physical calculator**.

Applications

- Office employees using laptops
- Call center agents
- Remote workers

7. IP-PBX Systems

Definition

IP-PBX systems manage **internal and external calls** within an organization using internet technology.

Examples

- 3CX
- Grandstream UCM Series
- Yeastar S-Series

Real-Life Analogy

Like a **school switchboard** that connects classrooms, offices, and the principal.

Applications

- Internal office calling
- Call forwarding and extensions
- Company-wide communication systems

8. VoIP Gateways

Definition

VoIP gateways connect **traditional telephone networks (PSTN)** with **VoIP networks**.

Examples

- Cisco VG Series Gateways

- Patton SmartNode Gateways

Real-Life Analogy

Like a **language translator** helping two people who speak different languages communicate.

Applications

- Connecting old phones to new VoIP systems
- Gradual migration from analog to VoIP
- Telecom infrastructure integration

9. Enterprise VoIP Solutions

Definition

Enterprise VoIP solutions are **advanced VoIP systems** designed for **large organizations** with complex communication needs.

Examples

- Avaya Aura
- Microsoft Teams Voice
- Mitel MiVoice

Real-Life Analogy

Like a **national school network** connecting many schools across the country.

Applications

- Large corporations
- Government institutions
- Multinational companies

10. Wholesale VoIP Services

Definition

Wholesale VoIP services are provided to **businesses that resell VoIP services** to customers.

Examples

- VoIP Innovations
- Twilio
- Nexmo (Vonage)

Real-Life Analogy

Like buying goods in **bulk** and selling them in smaller quantities.

Applications

- Telecom companies
- Internet service providers
- Call service resellers

11. SIP Trunking Services

Definition

SIP Trunking allows businesses to **connect their IP-PBX to external telephone networks** over the internet.

Examples

- Twilio Elastic SIP Trunking
- Flowroute
- Bandwidth

Real-Life Analogy

Like a **main road** connecting a school to the city.

Applications

- Making external calls
- Reducing phone line costs
- Scalable business communication

12. Open-Source VoIP Systems

Definition

Open-source VoIP systems have **free and editable source code**, allowing customization.

Examples

- Asterisk
- FreeSWITCH
- Kamailio

Real-Life Analogy

Like a **recipe you can change** instead of buying ready-made food.

Applications

- Learning and training
- Custom VoIP solutions
- Research and development
- Low-cost deployments

Application Area / Uses of VoIP

VoIP is widely used in **personal, business, and specialized communication**:

- **Personal Communication:**
 - Video/audio calls between individuals using apps.
 - Examples: WhatsApp, Skype, Viber.
- **Business Communication:**
 - Internal calls (extensions, inter-department).
 - Conference calls and video meetings.
 - Customer service centers (call centers).
 - Examples: RingCentral, Avaya IP Office, Microsoft Teams.
- **Mobile & Remote Communication:**
 - Use smartphones for international calls without high charges.
 - Telecommuting solutions for remote workers.
- **Specialized Areas:**
 - Unified communications (voice, video, messaging, collaboration).
 - Integration with CRM systems.
 - Emergency services using IP networks.

❖ Benefits of VoIP

VoIP has **many advantages over traditional phone lines**:

1. **Cost-effective:**
 - Cheaper calls, especially long-distance or international.
2. **Flexibility & Mobility:**
 - Make calls from anywhere with internet access.

- Use smartphones, laptops, or IP phones.
- 3. **Advanced Features:**
 - Call forwarding, voicemail-to-email, conferencing, video calls.
- 4. **Scalability:**
 - Easy to add/remove users in business systems.
- 5. **Integration with Other Applications:**
 - Works with CRM, email, chat, and collaboration tools.
- 6. **Unified Communication:**
 - Voice, video, and messaging in one platform.

❖ **Working Principle of VoIP**

VoIP converts **voice signals into digital data** and transmits them over the internet instead of traditional telephone lines.

Step-by-step process:

1. **Voice Capture:**
 - Microphone captures your voice as an analog signal.
2. **Analog-to-Digital Conversion:**
 - The VoIP system converts analog voice into digital packets using codecs (e.g., G.711, G.729).
3. **Packet Transmission:**
 - Digital voice data is split into packets and sent over the internet using IP (Internet Protocol).
4. **Routing:**
 - Packets travel via routers and networks to the destination.
5. **Digital-to-Analog Conversion:**
 - At the receiver, packets are reassembled and converted back to analog sound.
6. **Playback:**
 - The recipient hears your voice in real-time.

Essentially, VoIP works like sending **voice emails in real-time over the internet**.

❖ **Key VoIP Elements / Components**

VoIP systems require **hardware and software components** to function:

1. **IP Phones / Softphones:**

- Hardware phones (Cisco, Yealink) or software apps on PC/mobile (Skype, Zoom).

2. **VoIP Server / IP PBX:**

- Manages calls, routes them to the correct destination, and handles features like voicemail and conferencing.
- Can be **On-Premise** or **Hosted**.

3. **Gateways:**

- Connect VoIP networks to traditional phone networks (PSTN) for hybrid communication.

4. **Codecs:**

- Compress and convert voice signals into digital data packets.
- Examples: G.711, G.729, Opus.

5. **Network Infrastructure:**

- Routers, switches, and internet connection capable of handling VoIP traffic.

6. **Optional Elements:**

- Firewalls, session border controllers (SBCs) for security.
- Unified communication software integrating chat, video, and voice.

VOIP SYSTEM DIAGRAMS

A **VoIP System Diagram** is a **visual representation** of how a VoIP system works—showing the **components**, their connections, and how **voice signals flow** through the network. These diagrams help understand the **working, architecture, and setup** of VoIP systems.

1. Purpose of a VoIP System Diagram

- Shows how VoIP components (phones, servers, gateways, network) interact.
- Helps in planning and troubleshooting a VoIP network.
- Visualizes **call flow** from sender to receiver.
- Differentiates between **On-Premise, Hosted, and Hybrid VoIP setups**.

2. Typical Components in a VoIP Diagram

1. **User Devices:**

- IP Phones, Smartphones, Laptops (Softphones).

2. **Network Infrastructure:**

- Routers, Switches, Firewalls, Internet.

3. **VoIP Server / IP PBX:**

- Manages call routing, voicemail, conferencing.

4. **Gateways:**

- Connects VoIP network to **traditional PSTN**.

5. **Codecs:**

- Digital voice encoding/decoding (e.g., G.711, G.729).

6. **Optional Services:**

- Call management software, unified communication platforms, session border controllers (SBCs).

VoIP DIAGRAMS SOFTWARES

1. Microsoft Visio

- **Type:** Desktop diagramming tool (Windows)
- **Best for:** Professional network & system diagrams
- **Features:**
 - Network stencils (IP phones, servers, routers, firewalls)
 - Layering, templates, linkable data diagrams
 - Integration with Microsoft 365
- **Good for:** Enterprise-level VoIP architecture & documentation

2. Lucidchart

- **Type:** Web-based diagram maker
- **Best for:** Collaborative, cloud-accessible diagrams
- **Features:**
 - Drag-and-drop network shapes
 - Team sharing & real-time editing
 - Import Visio files
- **Good for:** Teams and remote work

3. draw.io / diagrams.net

- **Type:** Free online diagram tool
- **Best for:** Quick network or VoIP flow diagrams
- **Features:**
 - Plenty of network and VoIP icons
 - Offline/online usage
 - Exports to PNG, SVG, PDF
- **Good for:** Students, budget projects, simple diagrams

4. Cisco Packet Tracer

- **Type:** Network simulation & diagramming tool
- **Best for:** Realistic network setups (incl. VoIP devices)
- **Features:**
 - Simulation of network traffic & VoIP protocols
 - Drag-and-drop routers, switches, IP phones
 - Real packet behavior simulation
- **Good for:** Networking learners & professionals testing VoIP setups

5. SolarWinds Network Topology Mapper

- **Type:** Automated network mapper
- **Best for:** Large network diagrams including VoIP endpoints
- **Features:**
 - Auto-discover devices & topology
 - Export diagrams to Visio
 - Detects IP phones, servers, routers
- **Good for:** IT teams managing real VoIP deployments

❖ **IDENTIFICATION OF VOIP SYSTEM INSTALLATION REQUIREMENTS**

➤ **VoIP System Installation Requirements**

1. Evaluation of Environment

Before installing a VoIP system, you need to assess the environment to ensure compatibility, reliability, and efficiency.

A. Physical Site Survey

- Check the building or office layout: rooms, floors, network closets, cabling pathways.
- Identify power sources and network points for IP phones and servers.
- Assess areas where wireless VoIP devices may be used.
- Evaluate network hardware locations (switches, routers, gateways).

B. System Usage

- Determine the number of users and devices.
- Identify call patterns:
 - Internal vs external calls
 - Call volume per hour/day
 - Conference call needs
- Decide on features needed: voicemail, call forwarding, call recording, video calls.

C. Provide Blueprint Design

- Create a **network & system layout diagram**:
 - Show placement of IP phones, PBX/server, routers, switches, gateways.
 - Indicate internet connection and PSTN integration points.
 - Include redundant paths or backup systems if necessary.
- Define **call flow**: how calls move from devices → PBX → internet/PSTN.

2. Tools

Tools required for installation, setup, and troubleshooting:

- **Hardware Tools:** Screwdrivers, crimping tools, cable testers, punch-down tools
- **Network Tools:** Wi-Fi analyzers, network cable testers, bandwidth monitors, IP scanners
- **Software Tools:**
 - VoIP configuration software (PBX management tools)
 - Network diagram software (Visio, Lucidchart, draw.io)
 - Monitoring and troubleshooting tools (Wireshark, SolarWinds)

3. Materials

1. Cabling: Ethernet Cables (Cat5e / Cat6 Recommended)

- **Description:** Network cables used to connect VoIP devices (IP phones, PBX, switches, routers).
- **Use in VoIP system:**
 - Transmit voice and data packets between devices.
 - Cat5e: Supports 1 Gbps speed, sufficient for small to medium networks.
 - Cat6: Supports up to 10 Gbps, recommended for high-performance or future-proof networks.
 - Ensures reliable call quality with minimal latency or packet loss.

2. Connectors: RJ-45, RJ-11 (If PSTN Integration Needed)

- **Description:** Modular connectors for network and telephone cables.
 - **RJ-45:** Standard for Ethernet (network) connections.
 - **RJ-11:** Standard for traditional analog phone lines (PSTN).
- **Use in VoIP system:**
 - RJ-45 connects IP phones, switches, and PBX.

- RJ-11 connects gateways or hybrid systems to the PSTN network.
- Ensures secure and stable physical connections.

3. Patch Panels for Structured Cabling

- **Description:** A panel with multiple ports that organizes and terminates network cables in a structured way.
- Use in VoIP system:
 - Provides centralized management of cables for IP phones, servers, and network devices.
 - Makes future changes, troubleshooting, and maintenance easier.
 - Improves organization and cable longevity.

4. Rack Mounts for Servers and Network Devices

- **Description:** Standardized racks used to mount servers, switches, gateways, and other networking equipment.
- **Use in VoIP system:**
 - Keeps equipment organized, secure, and ventilated.
 - Supports structured cabling and easy access.
 - Facilitates cooling, maintenance, and expansion in enterprise environments.

5. Labeling Materials for Ports and Cables

- **Description:** Labels, tags, or color-coded markers used to identify cables, ports, and devices.
- **Use in VoIP system:**
 - Simplifies installation, troubleshooting, and maintenance.
 - Reduces errors in connecting devices.
 - Makes documentation and network management more professional.

4. Equipment

1. IP Phones (Desk Phones, Softphones, Mobile Clients)

- **Description:** Devices that users interact with to make or receive VoIP calls.
 - **Desk phones:** Physical phones connected via Ethernet or Wi-Fi.
 - **Softphones:** Software-based phones on laptops or desktops.



- **Use in VoIP system:**
 - Enables **voice communication over the internet**.
 - Provides **features like call transfer, voicemail, conferencing, and video calls**.
 - Can connect directly to **PBX or hosted VoIP service**.

2. VoIP Server / IP PBX (On-Premise or Cloud Subscription)

- **Description:** Central server that **manages and routes all VoIP calls**.
 - **On-Premise PBX:** Installed locally on company servers.
 - **Cloud PBX / Hosted VoIP:** Managed by service provider; accessed via internet.
- **Use in VoIP system:**
 - Handles **call routing**, voicemail, extensions, conferencing.
 - Provides **centralized management for all devices**.
 - Integrates with **SIP trunks** to connect external phone networks.
 - Supports **advanced features** like call recording, auto-attendants, and call queues.



3. Gateways for PSTN Integration

- **Description:** Hardware or software interface connecting **VoIP network to traditional phone networks (PSTN)**.
- **Use in VoIP system:**
 - Allows VoIP users to call landlines and mobile numbers.
 - Converts **digital VoIP signals into analog PSTN signals** and vice versa.
 - Ensures **compatibility between VoIP and legacy telephony systems**.



4. Network Switches & Routers (PoE switches recommended for IP phones)

- **Network Switches:** Connect all IP devices within a LAN, including IP phones and servers.
- **PoE (Power over Ethernet) Switches:** Supply **both power and data** to IP phones via a single Ethernet cable.
- **Routers:** Connect the LAN to the **internet**, handling external calls or cloud VoIP.
- **Use in VoIP system:**
 - Ensures **reliable connectivity for all VoIP devices**.
 - Manages **network traffic, QoS (Quality of Service), and VLANs** for voice prioritization.
 - PoE switches reduce **need for separate power sources** for IP phones.



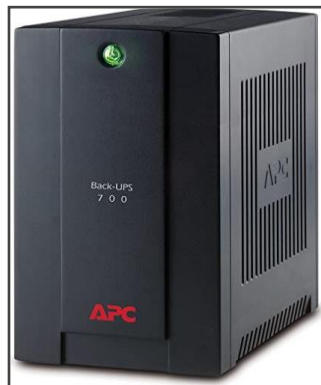
5. Firewall & Security Devices (for Network Protection)

- **Description:** Devices or software that **protect VoIP network from unauthorized access and attacks**.
- **Use in VoIP system:**
 - Protects against **VoIP hacking, eavesdropping, spam, and DoS attacks**.
 - Secures **SIP sessions and RTP voice streams**.
 - Can include **Session Border Controllers (SBCs)** for secure call handling between networks.



6. Uninterruptible Power Supply (UPS) for Critical Devices

- **Description:** Backup power system that provides electricity **when primary power fails**.
- **Use in VoIP system:**
 - Keeps **IP phones, PBX, switches, routers, and gateways** running during power outages.
 - Prevents **call drops and system downtime**.
 - Ensures **continuous operation of VoIP services**, especially for businesses and emergency lines.



5. VoIP Services

- **Hosted/Cloud VoIP** or On-Premise PBX service
- **SIP Trunks** for connecting to PSTN or other VoIP networks
- **Voicemail / Unified Messaging services**
- **Call forwarding, conferencing, video calls, mobile VoIP apps**

6. VoIP Network Requirements

- **Bandwidth:** Ensure sufficient internet speed and QoS (Quality of Service) for voice traffic
QoS (Quality of Service) is a set of network rules and technologies that **make sure voice calls get priority on the network** so that VoIP calls are **clear, smooth, and reliable**.

👉 In VoIP systems, QoS ensures that **voice traffic is delivered faster and more reliably than other data** like downloads or video streaming.

VoIP calls are **real-time**. Unlike emails or file downloads, **voice cannot wait**.

Without QoS, VoIP calls may suffer from:

- Delays
- Echo
- Broken or robotic voice
- Call drops

📌 That's why QoS is **mandatory** in VoIP system installations.

- **Network Design:**
 - Segmentation of voice traffic from data traffic
 - VLAN for VoIP devices recommended
 - **Latency & Jitter:** Must be minimal to avoid call drops or poor quality
 - **Redundancy:** Backup links or alternative routes for high reliability
 - **Security:** Encryption (SIP/TLS), firewall rules, and intrusion prevention.
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1. Design of VoIP System Diagram

➤ Design of VoIP system diagram

A **VoIP system diagram** visually represents how the VoIP system works. There are two main types:

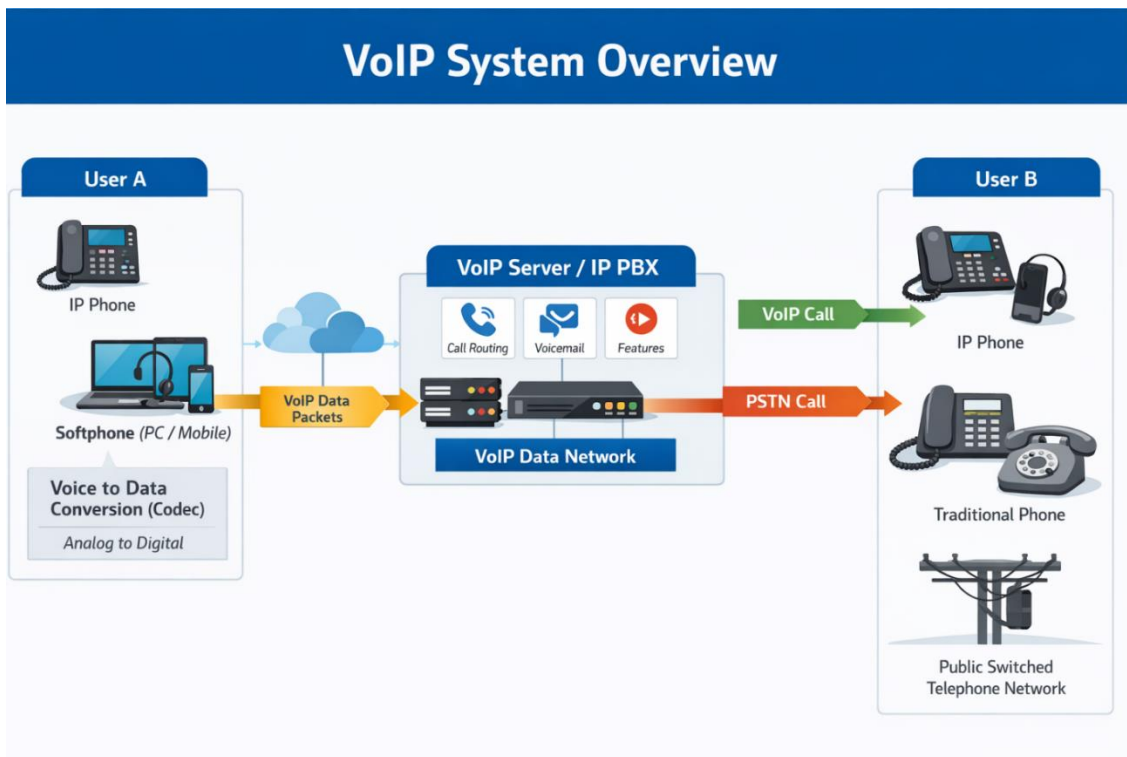
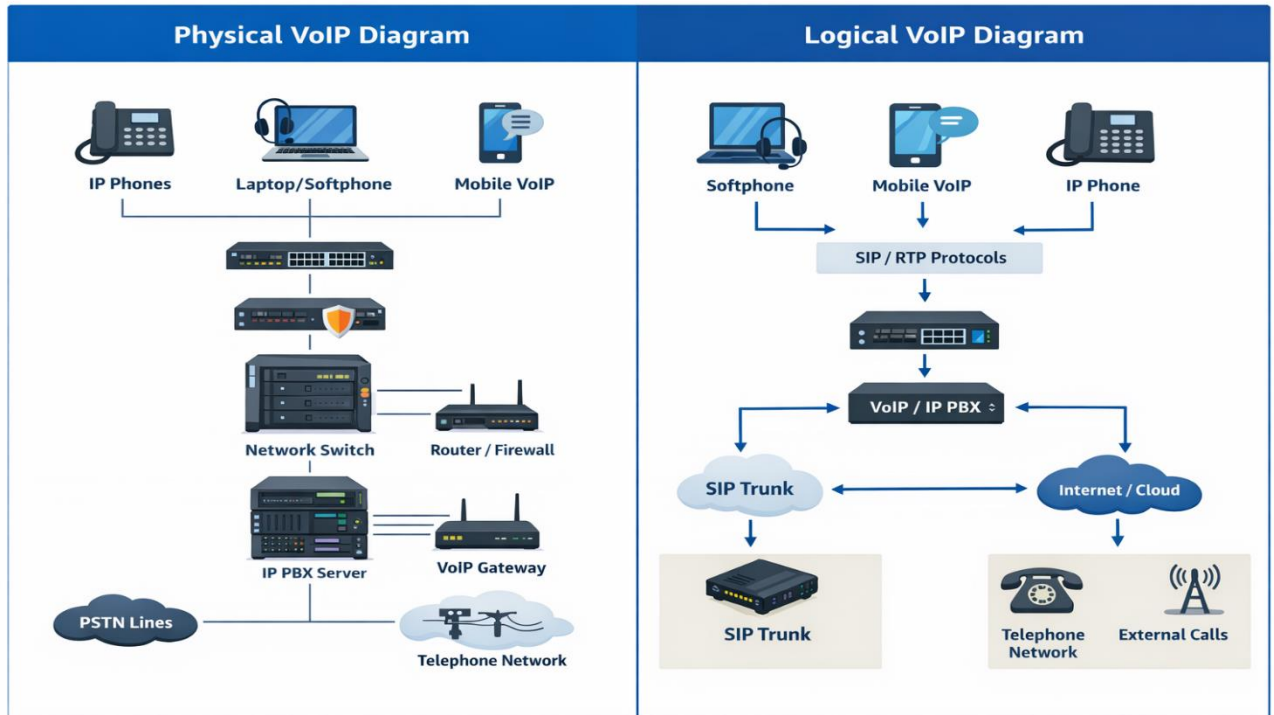
A. Physical Diagram

- **Definition:** Shows the **actual hardware and physical connections** in the VoIP network.
- **Purpose:** Helps installers and IT teams understand **where devices are located, how they are connected, and what equipment is used.**
- **Components usually shown:**
 - IP phones, softphones, mobile clients
 - IP PBX / VoIP server
 - Routers, switches, firewalls
 - Gateways to PSTN
 - Cabling (Ethernet, fiber)
 - Internet connection
- **Description / Key Points:**
 - Focuses on **physical placement and wiring.**
 - Shows **real devices, racks, closets, patch panels, PoE switches.**
 - Useful for **installation, maintenance, and troubleshooting.**

B. Logical Diagram

- **Definition:** Shows **how data and calls flow logically**, without focusing on physical hardware placement.
- **Purpose:** Helps network engineers and planners understand **call routing, protocols, and network design.**
- **Components usually shown:**
 - VoIP servers / PBX
 - SIP trunks / gateways
 - VoIP clients (phones, softphones, mobile apps)
 - Internet / LAN / WAN connections
 - Call flow paths (from sender → server → receiver)
- **Description / Key Points:**
 - Focuses on **communication flow and connectivity** rather than exact device location.
 - Highlights **call routing, firewall rules, VLANs, QoS for voice traffic.**

- Useful for **network design, troubleshooting, and VoIP optimization.**



❖ Use of Design Tools

Design tools are software or applications used to **visualize, plan, and document** VoIP networks, including physical layout, logical call flows, and system architecture. They help **reduce errors, improve communication, and support implementation and troubleshooting.**

1. Planning & Visualization

- **Purpose:** Design tools allow you to create a **visual blueprint** of your VoIP system before installation.
- **Uses:**
 - Map **physical locations** of IP phones, routers, switches, servers, and gateways.
 - Show **logical call flows**, including SIP trunks, IP PBX routing, and internet/PSTN connections.
 - Visualize **network segmentation**, VLANs, QoS paths, and security layers.
- **Benefit:** Helps IT teams and stakeholders understand the network structure clearly.

2. Simulation & Testing

- Some design tools, especially network simulators, allow **testing before deployment.**
- **Uses:**
 - Simulate **voice traffic and bandwidth requirements.**
 - Test **call quality under load.**
 - Detect potential bottlenecks, latency, jitter, or packet loss.
- **Tools for this:** Cisco Packet Tracer, GNS3

3. Documentation & Communication

- **Purpose:** Design tools create **official documentation** for IT, management, or client presentation.
- **Uses:**
 - Maintain **network diagrams** for troubleshooting and upgrades.
 - Share **installation plans** with technicians.
 - Provide **reference for VoIP service providers** for SIP trunk integration or support.
- **Tools for this:** Microsoft Visio, Lucidchart, draw.io

4. Collaboration & Updates

- Modern tools enable **team collaboration** and easy updates.
- **Uses:**
 - Multiple users can edit diagrams in real-time.
 - Changes to topology, devices, or call flow can be instantly shared.
 - Cloud-based tools allow remote teams to participate in planning.
- **Tools for this:** Lucidchart, Gliffy, draw.io

5. Standardization & Professional Presentation

- Design tools provide **professional icons and templates** for VoIP and network devices.
- **Uses:**

- Use standard symbols for IP phones, PBX servers, gateways, routers, and internet clouds.
- Present diagrams clearly for **management approvals or training purposes**.
- **Tools for this:** Microsoft Visio, OmniGraffle, Lucidchart

❖ Estimation of VoIP System Cost

Estimating the cost involves three main steps: **identifying item specifications**, **identifying item costs**, and **calculating the total cost**.

1. Identify Item Specifications

Determine **what equipment, materials, services, and infrastructure** you need for your VoIP deployment.

Example Specifications:

Item	Specification
IP Phones	Desk phone, SIP-enabled, PoE support, 50 units
Softphones / Mobile Clients	VoIP app licenses for 50 users
VoIP Server / IP PBX	On-premise server or cloud subscription, 1 server
Network Switches	24-port PoE switches, 2 units
Routers	High-speed router with QoS support, 1 unit
Gateways	PSTN gateway for 2 lines, 1 unit
Cabling	Cat6 Ethernet cable, 500 meters
Connectors	RJ-45, RJ-11, 100 pieces each
Patch Panels	24-port, 2 units
Rack Mounts	42U server rack, 1 unit
Firewall / Security	Hardware firewall with VoIP support, 1 unit
UPS	2000VA for critical devices, 2 units
VoIP Services	SIP trunks, cloud PBX subscription, per month

2. Identify Item Cost

Next, **find the unit cost** for each item. Costs can vary by region, brand, or service provider.

Example Item Costs Table:

Item	Unit Cost (USD)	Quantity	Total Cost (USD)
------	-----------------	----------	------------------

IP Phones	100	50	5,000
Softphones / Mobile licenses	5	50	250
VoIP Server / PBX	1,500	1	1,500
Network Switches	400	2	800
Routers	300	1	300
Gateways	250	1	250
Cabling	0.5 per meter	500	250
Connectors	1	200	200
Patch Panels	100	2	200
Rack Mounts	500	1	500
Firewall / Security	600	1	600
UPS	300	2	600
VoIP Services	20 per month	12 months	240

3. Calculate the Total Cost

- Sum up all the **total costs for equipment, materials, and services**.
- Include **installation labor, training, and maintenance** if applicable.

Example Calculation:

Equipment + Materials Total:

$5,000 + 250 + 1,500 + 800 + 300 + 250 + 250 + 200 + 200 + 500 + 600 + 600 = \mathbf{10,450 \text{ USD}}$

Services Total (1 year):

VoIP service: 240 USD

Grand Total:

$10,450 + 240 = \mathbf{10,690 \text{ USD}}$

Optional: Include **10–15% contingency** for unexpected costs:

$10,690 \times 10\% = 1,069 \rightarrow \text{Total Budget} \approx \mathbf{11,759 \text{ USD}}$

LEARNING OUTCOME 2: DEPLOY VOIP SYSTEM EQUIPMENT | Learning hours:10

❖ Selection of VoIP system Tools, materials and equipment

Tools are items used by technicians to **perform installation, configuration, testing, or maintenance tasks**.

They **do not become part of the VoIP system** after installation.

Key Characteristics

- Used **temporarily**
- Assist in **cutting, fixing, testing, or configuring**
- Removed after work is completed
- **Do not carry voice or data**

Examples of Tools (VoIP System)

- **Cable cutter** – Cut network cables
- **Wire stripper** – Remove cable insulation
- **Punch-down tool** – Fix wires into patch panels
- **Screwdrivers** – Tighten or loosen screws
- **Drilling machine** – Make holes for mounting
- **Crimping tool** – Attach RJ45 connectors
- **Network cable tester** – Check cable connection
- **Multimeter** – Measure voltage & continuity



Role in VoIP Deployment

- Preparing cables
 - Installing racks and devices
 - Terminating connectors
 - Testing connectivity and signal quality
-

2. Materials

Definition

Materials are **consumable or passive items** that are used during installation and **remain as part of the VoIP infrastructure**.

Key Characteristics

- Become **part of the installed system**
- Usually **passive** (no processing or intelligence)
- May wear out over time
- Support connectivity and organization

Examples of Materials (VoIP System)

- **Ethernet cables (Cat5e, Cat6)** – Carry network/VoIP data
- **Fiber optic cables** – High-speed, long-distance data transmission
- **RJ-45 and RJ-11 connectors** – Connect network or phone cables
- **Patch cords** – Connect devices to patch panels or switches
- **Jumper cables** – Short connections between network points
- **Patch panels** – Organize and manage multiple network cables
- **Cable trays and trunking** – Hold and protect cables neatly
- **Labels and tags** – Identify cables for easy maintenance

Examples of Materials (VoIP System) and Their Use

Ethernet cables (Cat5e, Cat6)

Carry network/VoIP data



Fiber optic cables

High-speed, long-distance data transmission



RJ-45 and RJ-11 connectors

Connect network or phone cables



Patch cords

Connect devices to patch panels or switches

Patch cords

Connect devices to patch panels or switches

Patch panels

Organize and manage multiple network cables



Jumper cables

Short connections between network points



Cable trays and trunking

Hold and protect cables neatly



Cable trays and trunking

Hold and protect cables neatly



Labels and tags

Identify cables for easy maintenance



Role in VoIP Deployment

- Transmitting voice and data
- Connecting devices
- Organizing and identifying cables

3. Equipment

Definition

Equipment refers to **active or powered devices** that **process, control, or manage VoIP communication**.

Key Characteristics

- Core components of the VoIP system
- Require **power supply**
- Perform **communication, routing, or switching**
- Remain operational throughout system life

Examples of Equipment (VoIP System)

- **IP Phones** – Make/receive VoIP calls
- **IP PBX / PBX Server** – Manage internal & external calls
- **VoIP Router** – Direct VoIP traffic on the network
- **VoIP Gateway** – Connects VoIP network to traditional phone lines
- **PoE Switch** – Supplies power and network to IP phones
- **Analog Telephone Adapter (ATA)** – Connect old phones to VoIP system
- **UPS (Power Backup)** – Keeps VoIP devices running during power loss
- **Channel Bank** – Converts multiple voice channels for VoIP or analog networks

Examples of Equipment (VoIP System) and Their Use

IP Phones <i>Make/receive VoIP calls</i>  <i>Make/receive VoIP calls</i>	IP PBX / PBX Server <i>Manage internal & external calls</i>  <i>Manage internal & external calls</i>	VoIP Router <i>Direct VoIP traffic on the network</i>  <i>Direct VoIP traffic on the network</i>	VoIP Gateway <i>Connects VoIP network to traditional phone lines</i>  <i>Connects VoIP network to traditional phone lines</i>
PoE Switch <i>Supplies power and network to IP phones</i>  <i>Supplies power and network to IP phones</i>	Analog Telephone Adapter (ATA) <i>Connect old phones to VoIP system</i>  <i>Connect old phones to VoIP system</i>	UPS (Power Backup) <i>Keeps VoIP devices running during power loss</i>  <i>Keeps VoIP devices running during power loss</i>	Channel Bank <i>Converts multiple voice channels for VoIP or analog networks</i>  <i>Channel Bank</i>

Role in VoIP Deployment

- Call processing and routing
- Voice signaling and media handling
- Network connectivity and power delivery

1. Selection of VoIP System Tools

These are the **tools needed for physical installation, cabling, and testing:**

Purpose: Ensures physical installation is **secure, accurate, and professionally done.**

Tool Type	Description / Use
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Cutting Tools	Cable cutters, wire strippers used to cut and prepare Ethernet cables and patch cords.
Drilling Tools	Drills for mounting devices, racks, or cable trays on walls or ceilings.
Fixing Tools	Screwdrivers, pliers, fasteners for mounting IP phones, switches, racks, and patch panels.
Patching Tools	Punch-down tools for terminating cables into patch panels, wall jacks, and connectors.
Testing Tools	Network testers, tone generators, multimeters, cable continuity testers for verifying connections and troubleshooting.

2. Selection of VoIP System Materials

These materials are essential for connecting and integrating the VoIP network:

Purpose: Materials ensure **proper connectivity, integration, and efficient network layout.**

Material	Description / Use
Connectors	RJ-45 for Ethernet, RJ-11 for PSTN lines. Connect devices to network or telephone lines.
Analog Telephone Adapter (ATA)	Converts analog phone lines to VoIP signals , enabling legacy phones to work with VoIP.
Cable	Ethernet cables (Cat5e/Cat6) for network connectivity; carry voice and data packets.
PoE Adapter	Provides Power over Ethernet for IP phones without separate power supply.
Patch Cord	Short Ethernet cables for connecting devices to patch panels or switches.
Jumper Cable	Used to connect patch panels to switches or routers within racks.

3. Selection of VoIP System Equipment

The core devices and hardware needed for a **functional VoIP system**:

 **Purpose:** Equipment forms the **core infrastructure** of a VoIP system, enabling users to communicate efficiently over IP networks.

Equipment	Description / Use
-----------	-------------------

Channel Bank	Converts multiple analog voice lines into digital signals for VoIP integration .
IP Phone	Physical phones connected to the network, used by end users to make/receive calls .
Power Backup (UPS)	Provides uninterrupted power for IP phones, PBX, switches, and gateways.
IP PBX Server / PBX Server	Central system managing call routing, extensions, voicemail, and conferencing .
VoIP Router	Connects LAN to the internet , manages traffic, ensures QoS for voice .
VoIP Gateway	Bridges VoIP network with PSTN or legacy phone lines , enabling external calls.
VoIP Headset	Used with softphones or IP phones for hands-free communication , improves comfort in offices or call centers.
PoE Switch	Supplies both power and data to IP phones, simplifying deployment and cable management.

❖ Perform VoIP system trunking

Trunking in VoIP refers to the **organized routing of network and voice cables** between equipment like IP PBX servers, switches, gateways, and IP phones. Proper trunking ensures **neat installation, protection of cables, and easy maintenance**.

1. Description of Trunking Installation Types

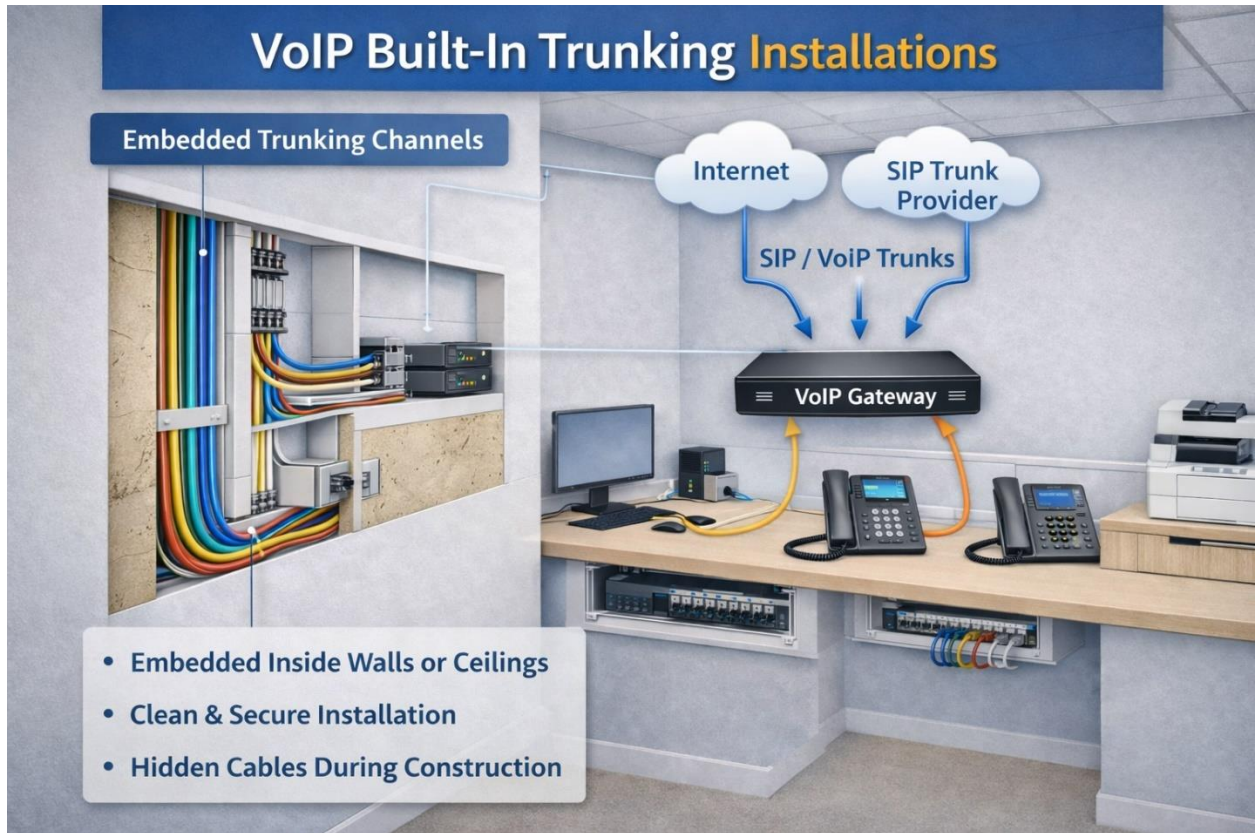
A. Visible Trunking

- **Description:** Cables run along walls or ceilings inside plastic or metal trunking channels, **visible to users**.
- **Advantages:**
 - Easy to install and modify.
 - Easy troubleshooting and maintenance.
- **Disadvantages:**
 - Aesthetic impact (not visually appealing).
 - Vulnerable to accidental damage.



B. Built-in Trunking

- **Description:** Trunking channels are **embedded inside walls or ceilings** during construction.
- **Advantages:**
 - Aesthetic – cables hidden.
 - Protected from physical damage.
- **Disadvantages:**
 - Harder to modify after installation.
 - Troubleshooting requires wall or ceiling access.

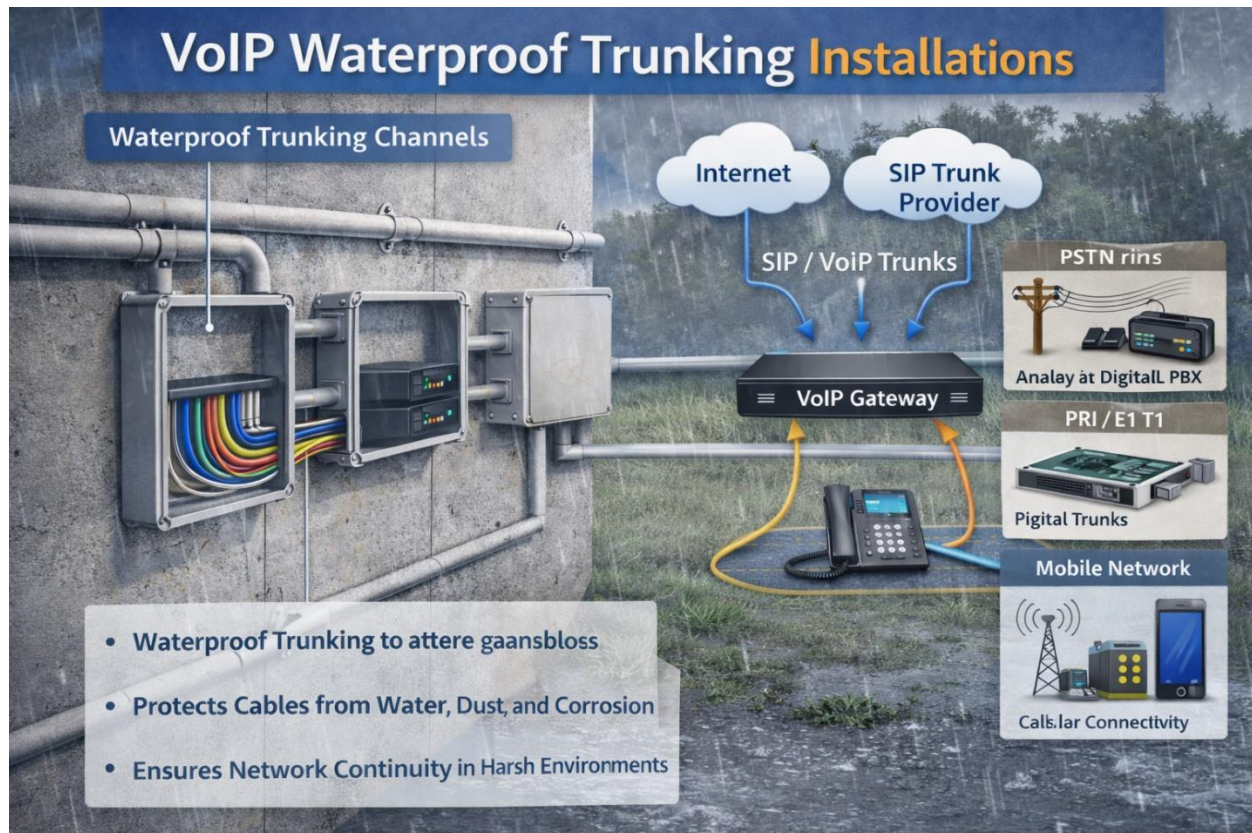


C. Semi-Built-in Trunking

- **Description:** Partially embedded; part of the trunking is visible.
- **Advantages:**
 - Moderate aesthetics and protection.
 - Easier to access than fully built-in.
- **Disadvantages:**
 - Compromise between aesthetics and accessibility.

D. Water Proof Trunking

- **Description:** Designed for areas **exposed to moisture or outdoor conditions**.
- **Advantages:**
 - Protects cables from water, dust, and corrosion.
 - Ensures network continuity in harsh environments.
- **Disadvantages:**
 - Higher cost.
 - Installation can be more difficult.



E. Underground Trunking

- **Description:** Cables buried underground inside ducts or conduits.
- **Advantages:**
 - Protects from vandalism and environmental hazards.
 - Invisible; does not occupy wall or ceiling space.
- **Disadvantages:**
 - Difficult to access for repairs.
 - High installation cost.

2. Description of Type of Trunking Materials

A. Plastic Trunking

- **Use:** Common for **indoor installations** and **lightweight cables** like VoIP Ethernet cables.
- **Advantages:**
 - Cost-effective.
 - Easy to cut, install, and modify.
- **Disadvantages:**
 - Less durable than metal.
 - Less protection against physical impact.

B. Stainless Steel Trunking

- **Use:** Used for **heavy-duty installations**, outdoor, or high-security areas.
- **Advantages:**

- Highly durable and impact-resistant.
- Fire-resistant and corrosion-resistant.
- **Disadvantages:**
 - Expensive.
 - Requires specialized tools and skills for installation.

3. Application of Cable Laying and Pulling Methods

Proper cable management ensures **network reliability, safety, and easy maintenance.**

A. General Method

- Pull cables along designed paths using **cable trays, conduits, or trunking.**
- Avoid sharp bends to prevent damage to Ethernet or fiber cables.

B. Cable Laying Arrangement

- Group similar types of cables together (voice, data, power).
- Maintain **minimum bend radius** for network cables.
- Avoid crossing power and network cables directly to reduce interference.

C. Segregation of Cables

- Separate **voice, data, and power cables** into different paths or conduits.
- Prevents **electromagnetic interference** affecting call quality.

D. Cable Marking

- Label cables at both ends and along the path using **tags, color codes, or numbers.**
 - Facilitates **troubleshooting, maintenance, and future expansion.**
-

4. Assemble VoIP Equipment

A. Rack Mount

- Place **PBX servers, switches, UPS, and gateways** into standard racks.
- Use **rack screws and rails** for secure mounting.
- Organize **cables neatly using patch panels and cable ties.**

B. IP PBX Server / PBX / PABX Server

- Install server in the **rack or designated equipment room.**
- Connect **Ethernet and power cables.**
- Configure for **call routing, extensions, SIP trunks, and VoIP services.**

C. VoIP Phones

- Connect IP phones to **PoE switches or network outlets.**
 - Configure via **DHCP or static IP**, register with PBX server.
 - For softphones or mobile clients, install **VoIP apps** and connect to PBX.
-

1. Advantages and Disadvantages of VoIP Trunking

Aspect	Advantages	Disadvantages
Visible Trunking	Easy access, easy to modify	Aesthetic issues, vulnerable
Built-in Trunking	Hidden, protected	Hard to modify or repair

Semi-built-in	Balanced access and aesthetics	Compromise, moderate protection
Water Proof	Moisture & dust protection	High cost, harder to install
Underground	Secure, invisible	High cost, difficult maintenance
Plastic Trunking	Cheap, easy to install	Low durability
Stainless Steel Trunking	Durable, fire/corrosion-resistant	Expensive, needs special tools

❖ **Connection of VoIP Devices**

Proper connection of VoIP devices ensures **reliable voice communication, minimal downtime, and maintainable infrastructure.**

1. Deployment of VoIP Devices

Steps:

1. Placement of Devices:

- Install **IP phones, softphones, mobile clients** at user desks or designated areas.
- Place **IP PBX servers, switches, gateways, UPS** in secure equipment racks.
- Maintain **accessibility for maintenance and ventilation.**

2. Network Connection:

- Connect IP phones to **PoE switches** or network outlets.
- Connect PBX server and VoIP gateways to switches/routers.

3. Configuration:

- Register devices with **PBX server** using SIP credentials.
- Assign **IP addresses** (DHCP or static).
- Configure **features** like voicemail, extensions, call forwarding, conferencing.

 **Tip:** Maintain proper **segregation of voice and data cables** to reduce interference.

2. Connection of Connectors

VoIP devices use **different types of connectors** depending on the medium:

Connector Type	Description	Use in VoIP Systems
Fiber Connector	Connects optical fiber cables (e.g., LC, SC, ST types)	For high-speed backbone connections between servers, PBX, or data centers
Coaxial Connector	Connects coaxial cables (e.g., RG6, BNC)	Legacy systems or specialized VoIP equipment integration

Twisted Pair Connector (RJ-45)	Connects Ethernet (Cat5e/Cat6) cables	Standard connection for IP phones, PoE switches, PBX servers
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Use high-quality connectors to avoid **signal loss or poor call quality**

3. Selection of Patching / Tagging Components

Organizing cables is crucial for network reliability and maintenance.

A. Patch Panel Types

- **Standard Patch Panels:** 24-port or 48-port, used for structured cabling.
- **Fiber Patch Panels:** For optical fiber connections.
- **Shielded Patch Panels:** For areas with electrical interference.

Use: Central termination points for cables, easier reconfiguration and troubleshooting.

B. Patch Cord Types

- **Straight-through Cables:** Connect devices to switches/patch panels.
 - **Crossover Cables:** Connect switches or routers directly (less common in VoIP).
 - **Fiber Patch Cables:** For connecting fiber ports on PBX or servers.
-

C. Jumper Cable Types

- **Short Ethernet Cables (0.5–3m):** Connect patch panels to switches or routers.
 - **Flexible Fiber Jumpers:** Connect fiber patch panels to transceivers.
 - **Purpose:** Reduce strain on cables, easy replacement, organized connections.
-

D. Patch Panel Management Techniques

- **Labeling:** Each port and cable should be clearly labeled.
- **Color Coding:** Assign colors to **voice, data, or power cables**.
- **Cable Management Accessories:** Use cable ties, Velcro straps, and cable ducts.
- **Rack Organization:** Maintain horizontal and vertical cable managers for neat routing.

 **Tip:** Proper patching and tagging **reduces downtime and simplifies troubleshooting.**

4. Testing of System Connectivity

After deployment, testing ensures **all devices and connections are working correctly**.

Testing Steps:

1. **Physical Testing:**
 - Use **cable testers** to check continuity, wiring correctness, and signal quality.
 - Test each patch cord, jumper, and connector.
2. **Network Testing:**
 - Ping IP phones, PBX servers, and gateways to confirm **network connectivity**.
 - Use tools like **Wireshark** to check for packet loss or jitter.
3. **VoIP Call Testing:**
 - Make **internal calls** between extensions.
 - Test **external calls** via SIP trunk or PSTN gateway.
 - Verify **features** like voicemail, conferencing, and call forwarding.
4. **Quality Testing:**
 - Measure **latency, jitter, and MOS (Mean Opinion Score)** for voice quality.
 - Ensure **QoS settings** prioritize voice packets.

LEARNING OUTCOME 3: CONFIGURE VOIP SYSTEM | LEARNING HOURS:30

❖ CONFIGURATION OF ROUTER-BASED VOIP SYSTEM (DETAILED NOTES)

A **router-based VoIP system** uses a network router as the **core device** to manage IP connectivity, route VoIP traffic, enforce Quality of Service (QoS), and provide secure access between VoIP devices, the IP PBX, and external networks (Internet / PSTN).

Proper router configuration is **critical** to ensure **call quality, security, and system reliability**.

1. VoIP Essential Router-Based Configurations

1.1 Router Interface Configuration

Definition

A router **interface** is a physical or logical connection point (Ethernet, fiber, WAN port) used to connect VoIP devices, LAN, and external networks.

Purpose in VoIP

- Connects **IP phones, IP PBX, switches, and VoIP gateways**
- Separates **LAN (internal VoIP network)** from **WAN (Internet / SIP trunk)**
- Enables proper routing of VoIP packets

Types of Interfaces in VoIP Routers

1. **LAN Interface**
 - Connects internal VoIP devices
 - Usually connected to PoE switches
2. **WAN Interface**
 - Connects to ISP / SIP trunk provider
3. **VLAN Interfaces (Logical)**
 - Separate **voice traffic from data traffic**
 - Improve performance and security

Typical Interface Configuration Tasks

- Assign **IP address and subnet mask**
- Enable the interface
- Configure VLANs (Voice VLAN)
- Apply QoS policies (voice priority)

Example (Conceptual)

LAN Interface: 192.168.10.1 /24 → VoIP Phones & PBX
WAN Interface: 41.x.x.x /30 → Internet / SIP Trunk

Importance

- ✓ Enables communication between VoIP devices
- ✓ Prevents IP conflicts
- ✓ Supports voice traffic prioritization

1.2 IP Default Gateway Configuration

Definition

The **default gateway** is the IP address of the router that devices use to send traffic **outside their local network**.

Role in VoIP Systems

- Allows IP phones and PBX to:
 - Reach **external VoIP services**
 - Access **SIP trunk providers**
 - Make international or PSTN calls
- Enables **internet-based VoIP calling**

How It Works

- When a VoIP device sends a packet to an unknown network:
 - The packet is forwarded to the **default gateway (router)**
 - Router forwards traffic to the WAN or Internet

Configuration Location

- On:
 - IP phones
 - IP PBX server
 - VoIP gateways
- Or automatically via **DHCP from router**

Example

Default Gateway: 192.168.10.1

(Router LAN IP address)

Impact if Misconfigured

- ✗ External calls fail
- ✗ SIP registration errors
- ✗ One-way or no audio

1.3 Password Configuration (Router Security)

Definition

Passwords protect the router from **unauthorized access**, preventing VoIP attacks and system compromise.

Why Passwords Are Critical in VoIP

- VoIP networks are frequent targets of:
 - Toll fraud
 - SIP hacking
 - Eavesdropping
- Weak router security can allow:
 - Call hijacking
 - Service disruption
 - Financial loss

Types of Router Passwords

1. Console Password

- Protects **physical access** to the router
- Prevents unauthorized configuration changes

2. Privileged (Enable) Password

- Required to enter **administrative mode**
- Controls full configuration access

3. Remote Access Password

- Used for:
 - SSH
 - Telnet (not recommended)
- Allows **secure remote router management**

Password Best Practices

- ✓ Use strong passwords (letters, numbers, symbols)
- ✓ Change default credentials immediately
- ✓ Use **SSH instead of Telnet**
- ✓ Limit management access by IP address

Security Impact

- Protects VoIP routing and call control
- Prevents malicious changes to SIP routing
- Ensures availability and reliability of VoIP services